Transformer-Based Cross-Domain Few-Shot Learning for Hyperspectral Target Detection

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***Abstract*— Deep learning-based methods have made significant progress in hyperspectral target detection (HTD). Unfortunately, limited target prior information and imbalance class resulting from the low occurrence probability of target leaves deep learning-based methods to confront bottlenecks. To ameliorate the abovementioned issues, a Transformer-based cross-domain few-shot learning (TCFSL) method is proposed for HTD. First, the TCFSL leverages cross-domain few-shot learning (FSL) to establish FSL tasks in both the source domain (SD) and the target domain (TD). This allows the TCFSL to learn transferable knowledge of the SD and distinguishable feature embedding model for the TD, to address the problems of target priori lacking and imbalance class. Second, feature-level and distribution-level domain adaptation (DA) is used to tackle the problem of domain shift in cross-domain FSL. The feature-level DA extracts intradomain information of the SD and TD to learn their common features to alleviate domain shift. The distribution-level DA based on cross-Transformer present interdomain distribution-level information aggregation and captures domain similarities of two data domains. By pursuing similarities between two data domains, the distribution-level DA block prompts specific FSL tasks in each domain, facilitating the target detection task. Finally, cross-domain FSL and DA blocks are trained in a unitary manner, which facilitates real-time information interaction and parameter adjustment between different blocks to achieve the optimal model. Experiments conducted on six HSI datasets indicate that the TCFSL outperforms 12 compared methods.**

***Index Terms*— Cross-domain few-shot learning (FSL), cross-Transformer, domain adaptation (DA), hyperspectral images (HSIs), target detection.**

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I. Introduction

A HYPERSPECTRAL image (HSI) is a 3-D cube that contains spectral and spatial information simultaneously [1]. Abundant spectral information contained in hundreds of adjacent spectral channels enables the HSI to distinguish different land covers that have different spectral curves due to their specific physical and chemical properties [2]. With the continuous development and optimization of current large-scale models, they have been applied in the field of remote sensing, such as SpectralGPT [3], which has made a breakthrough in the field of processing remote sensing data and the newly proposed SkySense [4], which can handle multimodal tasks. Therefore, HSIs are involved in civilian and military with multiple application aspects, like hyperspectral image classification [5], [6], hyperspectral change detection [7], and hyperspectral target detection (HTD) [8]. As a vital part of hyperspectral remote sensing, HTD is the task of finding and locating targets in HSIs with a given target reference spectrum [9]. The reference spectrum is usually obtained from the spectral library or the identified target pixels in the scene [10].

During the past few decades, HTD algorithms have been exploited extensively [11]. Original and simple algorithms for HTD, like Euclidean distance (ED) [12] and spectral angle mapper (SAM) [13], [14], are directly based on spectral similarity. Such methods are simple and effective, while their detection accuracy would decline in real scenarios due to the spectral variation and complexity of land covers. Alternatively, algorithms based on statistical theory were also proposed, like constrained energy minimization (CEM) [15], adaptive cosine/coherent estimator (ACE) [16], and spectral matched filter (SMF) [17]. Orthogonal subspace projection (OSP) [18], least-squares OSP (LSOSP) [19], and generalized likelihood ratio test (GLRT) [20] were also proposed and developed. Amounts of improved algorithms were developed rapidly, like the SMF derived from a subpixel mixing model (SPSMF) [21], the normalized likelihood ration test (NLRT) [22], the hierarchical CEM (hCEM) [23] and the ensemble-based cascaded CEM (E-CEM) [24], the background annihilated target-constrained interference-minimized filter (BA-TCIMF) [25], the chessboard-shaped topology for HTD (CTTD) [26],

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a joint optimization of super-resolution and target detection in hyperspectral images (HySRTD) [27], and so on. Chang [28] recently found that iterative kernel CEM (IKCEM) has even better performance. Most of these traditional methods and their improved versions were based on linear or simple nonlinear correlation to measure the similarity, which encountered bottlenecks in real detection scenarios. As an HSI always contains a variety of coverings with different spectral properties, these traditional methods are not good at mining complicated correlations between targets and backgrounds existing in an HSI.

Algorithms relying on representation theory were also proposed for HTD, which do not need any explicit assumption about the spectra distribution of an HSI. The sparsity-based target detector (STD) [29] and the combined sparse and collaborative representation (CSCR)-based [30] are two typical methods based on representation theory, which have been widely used as baseline algorithms in many conditions. Shen et al. [31] proposed a dual-sparsity approach (DSC) for HTD, which puts restraints on both the background and the target to obtain pure dictionaries. Though representation-based methods promoted HTD to a certain extent, their performances relied on dictionary constructing, and some background material signatures would be left out in a learned dictionary. More significantly, the raw features in an HSI may not be useful enough for HTD, and feature tuning that can learn more distinctive features is necessary.

In recent years, deep learning has made great progress in hyperspectral remote processing, due to their outstanding competence in extracting subtle and abstract features from amounts of redundant and complex data [32], [33], [34]. Li et al. [35] proposed a transfer learning method based on 1-D CNN for HTD (CNNTD). It used a labeled reference dataset to train a CNN, which was then used for the target detection of the testing HSI. Zhang et al. [36] proposed a more reasonable method called HTD-Net. The HTD-Net constructed a framework of similarity discrimination using the CNN (denoted as SD-CNN) and designed a sample augmentation strategy to overcome the lack of training samples in target detection. Zhu et al. [37] introduced a two-stream convolutional network-based target detector (TSCNTD). The TSCNTD took original pixel pairs as inputs instead of the difference of pixels to prompt the explicit comparison. It put forward a novel synthesis method to generate sufficient target samples with a target priori and some typical background pixels. Feng et al. [38] devised an attention-based multiscale spectral–spatial detector (AMSSD). It combined the 1-D CNN with a vision Transformer (ViT) to extract spectral and spatial features more accurately and designed a multiscale difference feature aggregation block to make the network obtain more discriminative representation competence. Shen et al. [39] proposed an interpretable representation network for hyperspectral target detection (HSI-IRN), which designed an interpretable representation network to extract nonlinear features of targets and background, and devised a lightweight multiscale Transformer to further enhance the nonlinear representation.

However, using deep learning techniques for HTD usually confronts two significant bottlenecks: 1) target priori in HTD are limited in most scenarios and 2) imbalance class due to the low emergence probability of targets compared with backgrounds. Limited target priori means insufficient training dataset and the class imbalance leads to underfitting or overfitting, deep learning-based models are challenging to meet the requirements in such cases. Although methods like HTD-Net, TSCNTD, and AMSSD attempt to address this issue by employing data augmentation, many of them rely on pixel-pair strategies, which can be both time-consuming and may lack robustness.

With amounts of transfer learning methods emerging in computer visual processing, it was also applied in hyperspectral remote sensing processing [40], [41], [42]. By utilizing labeled data from other data domains, transfer learning-based methods are promising in addressing the problem of limited priori in HTD. Until now, transfer learning-based methods have been widely used in hyperspectral classification, while related research is few in HTD [43], [44]. The CNNTD is an early attempt at transfer learning on HTD. However, transfer learning typically requires extensive network training to achieve better generalization effects; otherwise, it may struggle to improve performance across domains effectively.

Based on meta-learning, few-shot learning (FSL) has developed rapidly and is widely used in HSI processing [45]. Meta-learning trains the network by constructing tasks on the labeled dataset and then migrating it to the current target detection task, which can effectively reduce the consumption of network training. Wang et al. [46] proposed a meta-learning-based HTD using a Siamese network (MLSN), which expected that the network would not require considerable retraining to generalize to unfamiliar application contexts. The idea of meta-learning is first introduced for HTD in the MLSN. Although FSL is effective in identifying new unseen classes using limited training samples, their detection competence declines when new unseen classes have deviations from the training set.

Facing the above problems, Shi et al. [47] proposed a novel semi-supervised domain adaptive few-shot learning (SDAFL) model to adaptively transfer similarity/dissimilarity measurement from the source domain (SD) with sufficient labeled samples to the TD in an adversarial manner, where source data and target data can be collected by different sensors, that is, sensor-independent. At the same time, HSI processing is often faced with the problem of domain shift caused by sensor nonlinearity, weather conditions, acquisition methods, and other factors, as well as the different distribution of different data domains. SDAFL uses domain adaptation (DA) to solve the problem of domain drift. The SDAFL combined FSL and DA to bridge the gap between the SD with a large amount of labeled data and the TD with very few labeled data.

Although the SDAFL applied DA in FSL to enhance the detection accuracy of HTD, it still has some shortcomings. First, it trains DA and FSL in separate stages. That is, the few-shot similarity–dissimilarity learning (FSDL) network is first trained, and the semi-supervised domain adaptive (SDA) network is then trained by fixing the trained FSDL. This training strategy will lead to the suboptimal model of the network. Second, SDAFL only focuses on the feature-level information between HSI patches but ignores the distribution-level

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information between different data domains. Finally, both MLSN and SDAFL only train FSL networks in the SD, and the trained framework lacks the pertinence of detection tasks in the TD.

Motivations: In the current research field of HTD, there are several significant challenges. The first and foremost is the scarcity of a priori information about the target, which, in most cases, leads to insufficient training datasets, which, in turn, seriously affects the generalization ability and detection accuracy of the model. In addition, the class imbalance problem between the target and the background further exacerbates this dilemma, making the model highly susceptible to favoring the identification of background features and neglecting the capture of target features during training, which leads to serious bias in the detection results. Although existing approaches such as HTDNet, TSCNTD, and AMSSD attempt to alleviate these problems through data augmentation, they rely on pixel-pairing strategies, an approach that not only consumes a large amount of computational resources but also exhibits low robustness in complex scenes.

In the face of these significant challenges, a Transformer-based cross-domain few-shot learning (TCFSL) method is proposed. This innovative approach aims to fundamentally solve the aforementioned issues. First, TCFSL constructs FSL tasks in both the SD and TD through cross-domain few-sample learning (FSL). This allows it to learn transferable knowledge in the SD and build distinguishable feature embedding models for the TD. This effectively addresses the problems of missing a priori information and class imbalance in the TD. Second, TCFSL tackles the domain bias issue in cross-domain FSL through feature-level and distribution-level DA techniques. Feature-level DA reduces domain bias by extracting intradomain information from the SD and TD and learning common features. Distribution-level DA, based on the cross-domain Transformer mechanism, aggregates distribution information across domains and captures similarities between the two data domains. Finally, by emphasizing the similarities between data domains, the distribution-level DA module enhances specific FSL tasks within each domain, thus improving the performance of the target detection task. Ultimately, TCFSL trains the cross-domain FSL and DA modules together, ensuring real-time information interaction and dynamic parameter adjustment among the modules for model optimization.

1) A cross-domain FSL framework TCFSL is proposed for HTD, which can effectively detect targets in HSIs. Specifically, TCFSL utilizes cross-domain FSL tasks to embed transferable information from the SD and discriminative features from the TD into the model. This approach achieves effective migration of cross-domain knowledge and significantly enhances the model's generalization capability.
2) A DA technology that combines two levels of scale of feature-level and distribution-level is proposed, which solves the problem of domain deviation in the process of domain transfer. Different from previous networks, TCFSL achieves the DA through two levels: feature-level and distribution-level adjustments. This approach enhances the adaptation process by aligning both the features and distributions across domains, ensuring improved model generalization from one domain to another.
3) TCFSL treats the FSL and DA blocks as a global optimization, rather than in stages, which solves the problems of suboptimal models and insufficient target DA of existing algorithms. This integration aims to enhance both the efficiency of FSL tasks within each data domain and the robustness of the DA across different domains simultaneously. This unified approach ensures that improvements in the FSL task acceleration also contribute to enhanced detection accuracy in target detection tasks, making it a holistic solution to the question posed by domain deviation during domain transfer processes, thereby obtaining a more optimal framework.

The rest of this article is arranged as follows. Section II introduces the proposed method in detail. Section III provides the experimental results. Section IV gives the conclusion.

II. Related Works

As a preliminary work and to better motivate our proposed method, expositions of some related concepts and primary principles are given in this section. First, the concept and relations among transfer-learning, meta-learning, and FSL are interpreted. Then, the important network blocks, ViT and the cross-ViT model are introduced in detail.

A. Transfer Learning, Meta-Learning, and FSL

Strictly, transfer learning and meta-learning are two application aspects of machine learning, respectively. Transfer learning aims to use the knowledge of the SD to help the learning of the TD, especially when the data of the TD is scarce or difficult to obtain, it can use the knowledge of the SD to improve the model performance through transfer learning. While meta-learning attempts to train the model with great generalization ability and can adapt to new tasks quickly by observing and learning the patterns and rules of multiple tasks. Namely, transfer learning focuses on solving problems such as data scarcity or domain shifting, while meta-learning focuses on learning how to learn.

FSL is the problem of learning when confronted with a small number of training samples. Generally, the FSL problem can be solved under the framework of meta-learning, so FSL can be regarded as a specific application of meta-learning. In other words, a few-shot is a kind of meta-learning that just uses several training samples. FSL involves learning transferable knowledge from a large amount of labeled data to generalize to new classification or detection tasks with few priori. In common few-shot setups, each FSL task T is made of support set S and a query set Q, where S is a labeled set and Q is an unlabeled set, with the network learning on the S and evaluating on the Q. All these support sets and query sets are sampled from the base set that is constructed from source/TD data. Suppose that S in a meta-task consists of C classes with K samples, then the FSL task is called a C-way K-shot task, denoted as an episode. In a meta-task, samples

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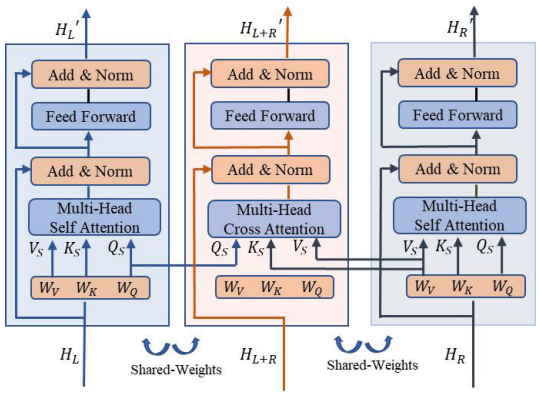


Fig. 1. Backbone of the ViT and the cross-Transformer. The left and the right branches are the structure of the ViT and the middle branch is the structure of the cross-Transformer.

of the query set Q and the support set S are sampled from the same classes, but they do not overlap each other.

In target detection, the obtained prior target information is limited, so a labeled dataset from another domain different from the TD (namely, the SD) is also used, so-called cross-domain FSL. The cross-domain FSL uses a large scale of labeled datasets from the SD and a certain number of labeled samples in the TD for network training so that it overcomes the problem of limited prior target information.

B. ViT and the Cross-Attention of It

ViT and cross-Transformer: the ViT consists of a multihead attention and a feedforward block and adopts additional learnable tokens to represent global features [48]. The ViT first crops an image into patches and sent into a linear projection layer and a position encoding is added. Embeddings after position coding are sent into the multihead attention layer. Multihead attention is composed of self-attention. In self-attention, every embedding is multiplied by three learnable matrices to obtain new vectors, called query vector, key vector, and value vector, all of which constitute Q, K, and V, respectively. Then, the self-attention is written as

$$\begin{aligned} Z &= \text{Attention}(Q, K, V) \\ &= \text{softmax}\left(\frac{Q \times K^T}{\sqrt{d}}\right) \times V \end{aligned} \tag{1}$$

where d is the number of dimensions of the query vector and the key vector.

To extract relevant features from different aspects validly, the multihead attention is devised, written as

$$\begin{aligned} \text{MultiHead}(Q, K, V) &= \text{Concat}(\text{head}_1, \ldots, \text{head}_h) \times W^O \\ \text{head}_i &= \text{Attention}(Q, K, V) \end{aligned} \tag{2}$$

where W^O is a learnable parameter matrix after connection and h is the number of heads in multihead attention.

The relationship of multiattention cross-attention in the ViT is shown in Fig. 1. In Fig. 1, H_L and H_R are the inputs of the ViT and output corresponding H'_L and H'_R, H_{L+R} is the input of cross-Transformer branch and outputs corresponding H'_{L+R}. In the cross-Transformer, its query is from the left branch, while the key and value are from the other branch. Such an operation attempts to pursue the correlation of the left and the right inputs, namely the similarity of two inputs.

The ViT aims to capture nonlocal features spanning a long-range, which is proper for global spatial feature extraction in $\mathcal{F}$. The cross-Transformer can capture relevant information of different scales, different distributions even different modalities, so it makes sense to use the cross-Transformer to extract distribution-level information and realize the interdomain attention, to achieve the final distribution-level DA.

III. Proposed TCFSL Method

As shown in Fig. 2, the TCFSL detector mainly consists of a feature extractor, two cross-Transformer blocks, and two domain classifiers. The TCFSL framework is divided into four parts: a training sample generation and feature extractor, a cross-domain FSL, and feature-level and distribution-level DA blocks.

A. Training Sample Generation and Feature Extraction

As only a small amount of target prior information can be obtained in the HTD task, a predetection is needed to choose several reliable background samples for the construction of the TD FSL task. To obtain pure and reliable background samples, the SPSMF [21], which performs well in suppressing background, is adopted as the coarse detector. As shown in Fig. 2, after obtaining the coarse detection map, several reliable background samples are randomly selected from the backgrounds of the coarse detection map. Then, the selected background pixels and prior target pixels are used as training samples for TD FSL tasks.

To ensure that samples from the SD and the TD have the same dimension before importing into the feature extractor, two mapping layers (denoted as M^s and M^t) are used for SD samples and TD samples, respectively. In the mapping layer, dimension mapping is implemented using a 2-D CNN. Suppose $I \epsilon R^{l \times l \times d}$ is the input cube from HSIs, where l denotes spatial patch size and d denotes the number of bands, then the output of mapping layer is $I' \epsilon R^{l \times l \times d_m}$. d_m is the number of bands after transformation. Feature extraction mainly concludes two parts: the feature extractor, denoted as F, and the cross-Transformer module.

Feature Extractor: The feature extractor F is to extract discriminative spectral–spatial features of both SD and TD samples. F mainly concludes two parts: A U-net framework and a ViT-based spatial feature extraction block, as shown in Fig. 3. The U-net takes the central pixel of the patch I' as input. It consists of an encoder and a decoder, with the 1-D CNN and 1-D transposed CNN as backbones, respectively. The encoder contains four downsampling stages with every stage composed of 2 or 3 convolutional layers and a max-pooling layer. Every convolutional layer is followed by batch normalization (BN) and dropout operations to prevent overfitting. The number of convolutional kernels for each stage increases accordingly, that is, 8, 16, 32, and 64. The decoder consists of four upsampling stages with every stage containing two or three transposed 1-D CNN (ConvTransposed1d) layers and a

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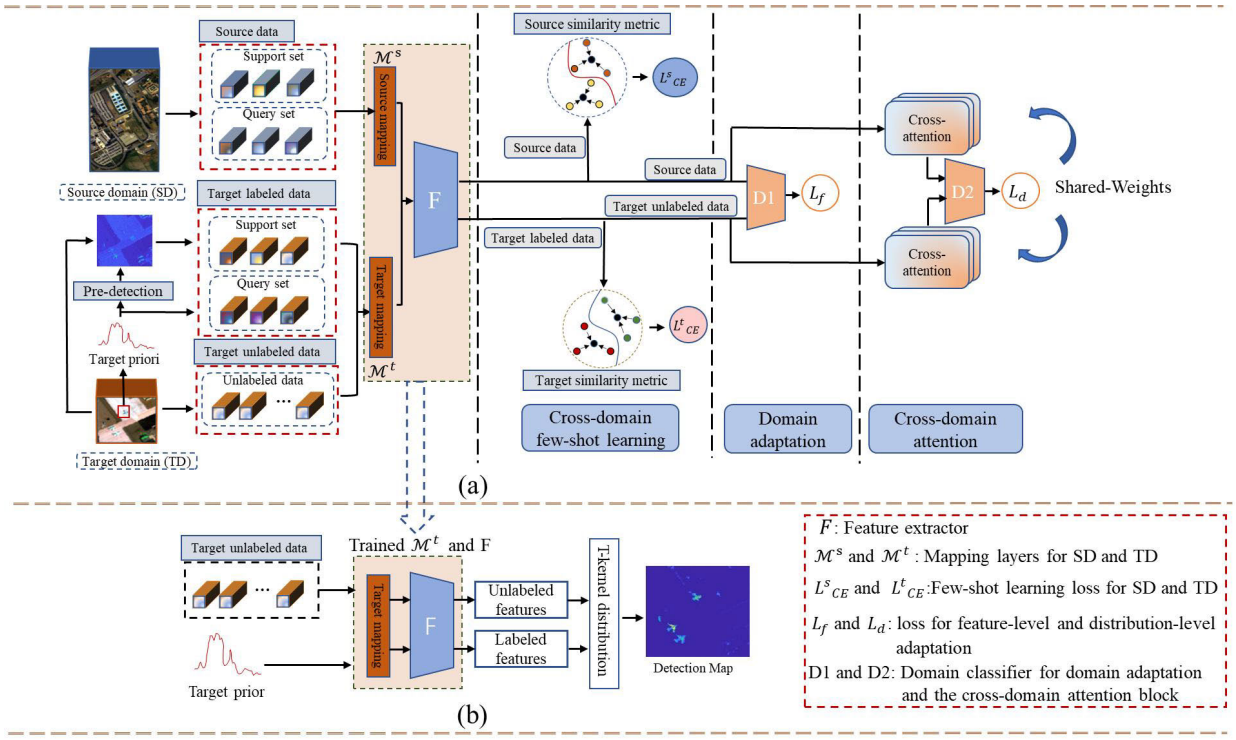


Fig. 2. Structure of the TCFSL framework. (a) Training phase and (b) testing phase. Three parts are contained in (a), namely the training sample generation and feature extractor, the cross-domain FSL, the DA, and cross-domain attention.

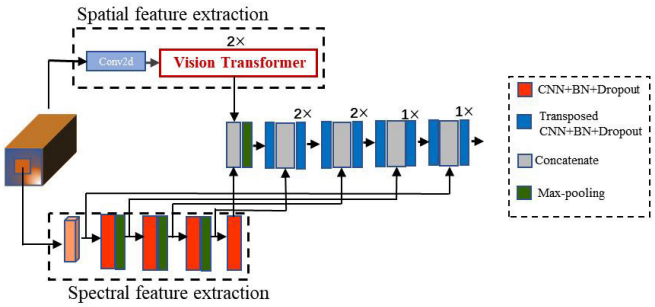


Fig. 3. Structure of the feature extractor.

max-pooling layer. The number of the first ConvTransposed1d kernels of every stage decreases regularly, that is, 64, 32, 16, and 8. The ViT-based block takes the patch I' as input for spatial information extraction. A 2-D CNN layer used is to process the patch to further aggregate spatial information before the ViT module. The output of the ViT-based spatial extraction branch is concatenated with the output of the convolutional layer in the last downsampling stage, then the concatenated embedding is through the max-pooling layer and sent to the decoder.

B. Cross-Domain FSL

In the TCFSL, there are two dataset resources, called the SD and TD. Therefore, there are two FSL tasks in such a scenario: the SD FSL task T_s and the TD FSL task T_t. In the SD, all data are labeled for FSL tasks. In the TD, only a few labeled samples T_t^{tr} are used, so-called few-shot dataset, and unlabeled data which are relatively larger than the few-shot dataset are used for testing task T_t^{te}. In SD episodic training, the SD support set $S_s = (x_n^s, y_n^s)_{n=1}^{C\times K}$ is composed of randomly selected C classes with K samples per class to form T_s. The SD query set has $Q_s = (x_m^s, y_m^s)_{m=C\times K}^{C\times K+T}$ samples (y_n^s and y_m^s denote corresponding class labels). In every episode, the support set S_s serves as the labeled training set on which the network is trained, then the query set Q is used as testing samples to minimize the loss function of its predictions. Episodic training for the TD is similar to that for the SD, the difference is that data augmentation is needed for TD few-shot data because only a few samples are given.

The flowchart of TCFSL is shown in Fig. 2, in which (a) and (b) diagrammatize the training phase and the testing phase, respectively. The TCFSL attempts to accomplish two kinds of FSL tasks: one FSL task for SD and the other one for TD. Tasks for the SD and TD are processed alternatively, to better seek for transformable knowledge in the source classes and learn a more discriminative embedding model to the target classes simultaneously. In FSL, the embedded features through F are input into FSL space to perform FSL task learning using the ED. Take FSL in the SD as an instance, in each episode, the support set S_s and a query set Q_s are sent into $F(M^s(\cdot))$ to obtain embedding features. Then, FSL is performed by calculating the ED between the embedding features of the query set and the prototype (the mean value of embedding features of the support set) of each class to minimize the prediction loss. The prediction probability of a query set sample $\mathbf{x}_k^s$ is calculated as

$$P\left(y_k^s|\mathbf{x}_k^s\right) = \text{Softmax}\left(-d\left(\mathbf{z}_k^s, \mathbf{z}_k^{\text{mean}(c)}\right)\right) \tag{3}$$

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where $d(\cdot)$ denotes an ED function. $k = 1, \ldots, N$, and N is the number of classes of support set. $\mathbf{z}_k^s$ is the embedded features of the query set. y_k^s is the true label. $\mathbf{z}_k^{\text{mean}(c)}$ is the mean value of the embedded features of the cth class support set, also called the prototype of the cth class ($c \in C_s$). The loss function is defined as the negative logarithmic probability of the true class k and its expectation loss can be expressed as

$$L_{\text{CE}}^s = E[L] = \frac{1}{N}\sum_{k=1}^{N} L_{\text{CE}}^s\left(y_k^s, P\left(y_k^s|\mathbf{x}_k^s\right)\right) \tag{4}$$

where $L_{\text{CE}}^s(y_k^s, P(y_k^s|\mathbf{x}_k^s))$ is the loss function for a single class, calculated as

$$\begin{aligned} &L_{\text{CE}}^s\left(y_k^s, P\left(y_k^s|\mathbf{x}_k^s\right)\right) \\ &= -\ln P\left(y_k^s|\mathbf{x}_k^s\right) \\ &= -\left\{y_k^s \ln P\left(y_k^s|\mathbf{x}_k^s\right) + \left(1-y_k^s\right)\ln\left(1-P\left(y_k^s|\mathbf{x}_k^s\right)\right)\right\}. \end{aligned} \tag{5}$$

FSL in the TD undergoes a similar procedure as the SD, and $L_{\text{CE}}{}^t$ is also calculated like (4). But noting that $L_{\text{CE}}{}^t$ is calculated on the basis of meta-tasks constructed by augmented data using T_t^{tr}.

Through FSL in both the SD and TD simultaneously, the cross-domain few-shot can not only learn captured transferred knowledge from the SD to facilitate target detection in the TD but also learn a reliable classification discrimination model for the TD. Besides, due to the number of samples in different categories being the same in each episode training of FSL, the issue of class imbalance is avoided.

C. Feature-Level DA Blocks

To make the embedding space learned on the SD data more adaptable to the TD which distributes differently, a DA block is devised. Related definitions are identified as follows: D_d and D_a are two domain discriminators for the DA and cross-domain attention. $P_s(x)$ and $P_t(x)$ are probability distributions on SD and TD data, respectively. T_u is the unlabeled data in the TD. g is the domain label of SD and TD data, in which 1 denotes SD data (g^s) and 0 denotes TD data (g^t).

The DA block is conducted on the embedded features of the SD and TD for each meta-task $f^{s/t}$ (s/t represents the SD or TD and f denotes features that derive from the $F(M^s(\cdot))$). After obtaining embedded features $f^{s/t}$, domain classifier D_f is used to calculate the domain classification probability of $f^{s/t}$. Among them, $P_s(x)$ and $P_t(x)$ are the predicted probabilities of SD data classification and TD data classification, respectively, and g^s and g^t are their labels. The binary cross-entropy loss with logit (BCEwithLogit) is used to calculate the loss between predicted probability $P_s(x)/P_t(x)$ and g. The loss function is calculated as

$$\begin{aligned} L_D &= -\frac{1}{N}\sum_{i=1}^{N}\left[g^s(i)\ln(\sigma(P_s(x_i))) + \left(1-g^s(i)\right)\ln(1-\sigma(P_s(x_i)))\right] \\ &\quad -\frac{1}{N}\sum_{i=1}^{N}\left[g^t(i)\ln(\sigma(P_t(x_i))) + \left(1-g^t(i)\right)\ln(1-\sigma(P_t(x_i)))\right] \end{aligned} \tag{6}$$

where $\sigma(\cdot)$ is a sigmoid function. This block is to characterize and aggregate intradomain nonlocal relationships among samples between SD and TD.

D. Distribution-Level DA Blocks

In the cross-domain FSL, class categories of the SD task should be much larger than the TD task, so an HSI for hyperspectral classification is always used for sample generation for SD FSL tasks. Due to different application characteristics, distributions of data used for hyperspectral classification and HTD are quite different. Specifically, a dataset for hyperspectral classification concludes multiple classes, which spread throughout the dataset widely, while a dataset for HTD only contains two classes, so-called background and target, where the target only occupies a small percentage and the background accounts for the majority with a variety of coverings. Although the feature-level DA is applied, intradomain feature-level information is not sufficient to alleviate the interference of domain deviation without the supplement of cross-domain distribution-level information. Thus, distribution-level DA is necessary when the SD and TD have heavy distribution deviation. Out of this consideration, a cross-Transformer is applied to aggregate interdomain information, and a distribution-level DA block is proposed. Related definitions are identified in Section III-C.

Considering that cross-attention in the Transformer is good at aligning different distributions, even from different modalities (e.g., vision-to-vision, vision-to-text, and text-to-speech) [49], and to capture the domain information at the distribution level, the cross-attention in the Transformer is applied to the SD and TD in the proposed framework, respectively, as shown in the right column in Fig. 2(a).

In SD episode training, $f^{s/t}$ from the SD and TD are equally divided into two pieces and fed into the cross-Transformer block, respectively. Then the cross-Transformer extracts nonlocal relation features in a larger range among samples of different coverings, to obtain the distribution-level information of two data domains, denoted as $f_d^{s/t}$. The $f_d^{s/t}$ contains relevant information among different pixel patches in the SD and the TD, that is, the nonlocal distribution-level information. Then, the domain discriminator D_a is used to obtain the domain class probability of $f_d^{s/t}$. The loss function L_A is written as

$$\begin{aligned} L_A &= -\frac{1}{N}\sum_{i=1}^{N}\left[g^s(i)\ln(\sigma(P_s(x_i))) + \left(1-g^s(i)\right)\ln(1-\sigma(P_s(x_i)))\right] \\ &\quad -\frac{1}{N}\sum_{i=1}^{N}\left[g^t(i)\ln(\sigma(P_t(x_i))) + \left(1-g^t(i)\right)\ln(1-\sigma(P_t(x_i)))\right] \end{aligned} \tag{7}$$

distribution-level DA in TD episode training is likely as that in the SD, the different is: In SD training, f_d^t is constructed by unlabeled data in the TD, while in TD training, f_d^t is constructed by the query set instead. By minimizing the loss, the cross-domain attention block seeks similarities between the

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SD and TD, to prompt classification accuracy in each specific domain.

Combined with FSL and DA blocks, the total loss of the TCFSL is defined as

$$L_{\text{total}} = L_{\text{CE}}^{s} + L_{\text{CE}}^{t} + \lambda_1 L_D + \lambda_2 L_A \tag{8}$$

where λ_1 and λ_2 are two regularization parameters that control the extent between feature-level and distribution-level DA.

E. Target Detection Based on T-Kernel Distribution

In the proposed TCFSL, the feature extractor and cross-ViT are simultaneously used to conduct episode training and domain alignment, and T^{tr} is augmented as the support set and the query set in the training phase. While in the testing phase, only the feature extractor is used, as shown in Fig. 2(b), and the t-distribution kernel is adopted to calculate the prediction results. The target detection via t-distribution kernel is defined as

$$r_{\text{pre}} = \frac{1}{1 + \| f_t^{\text{te}} - f_t^{\text{tgt}} \|_2^2} \tag{9}$$

where r_{pre} is the detection map and f_t^{tgt} and f_t^{te} denote the prior target signature and testing sample signature through the trained $F(\cdot)$, respectively. $\| \cdot \|_2^2$ is the ED. The t-kernel distribution converts embedding features of testing pixels into corresponding prediction probabilities, which simplifies the acquisition of prediction results.

IV. Experiments

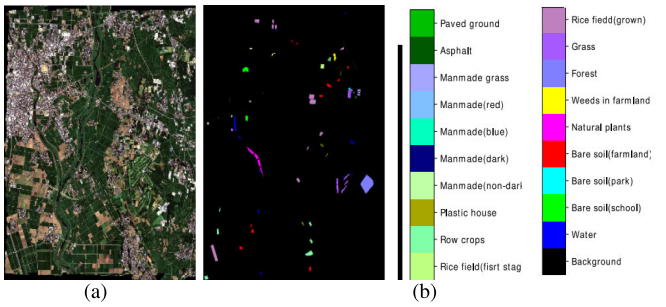
In this section, we apply the proposed TCFSL method on six real hyperspectral datasets and compare it with twelve classic and state-of-the-art methods to evaluate its performance.

A. Experimental Datasets

In some scenarios, HSIs in the SD and TD may come from different sensors. Most available public HSIs for experiments of HTD are captured by the Airborne Visible/InfraRed Imaging Spectrometer (AVIRIS) sensor. Therefore, the SD dataset is selected from the other sensors different from AVIRIS. Details of selected HSIs are presented as follows.

1) *Source Data Chikusei:* This airborne hyperspectral data was captured in Chikusei, Japan, on July 29, 2014, by the Headwall Hyperspec-Vnir-C sensor [50]. The Chikusei dataset has 128 bands ranging from 363 to 1018 nm in the spectra, and it consists of 2517 × 2335 pixels with a spatial resolution of 2.5 m. Nineteen classes are contained in the data, including urban and rural areas. The pseudo-color image and ground truth are shown in Fig. 4(a) and (b), respectively.
2) *Target Data:*

a) *IM1:* This image was captured on Gulfport with a spatial resolution of 3.4 m by the AVIRIS sensor. It is a scene of the airport of the dataset Airport-Beach-Urban (ABU) [51] and has a size of 100 × 100 × 191, in which three airplanes are regarded as targets. The false-color image and ground truth of IM1 are shown in Fig. 5(a) and (f), respectively. The IM1 has a size of 100 × 100 × 191 and is captured on the Texas Coast with a spatial resolution of 17.2 m.



(a) (b)

Fig. 4. Source data Chikusei. (a) False-color image. (b) Ground-truth map.

b) *IM2:* This image has a size of 100 × 100 × 193 and is captured on San Diego with a spatial resolution of 7.5 m. It is a scene of the beach of the ABU dataset and is also captured by the AVIRIS sensor. The false-color image and ground truth of IM2 are shown in Fig. 5(b) and (g), respectively.

c) *IM3:* This image has a size of 100 × 100 × 191 and is captured on the Texas Coast with a spatial resolution of 17.2 m. It is a scene of urban of the ABU dataset and is captured by the AVIRIS sensor. The false-color image and ground truth of IM3 are shown in Fig. 5(c) and (h), respectively.

d) *IM4:* This image has a size of 64 × 64 × 276, where 64 × 64 is spatial size and 276 is the number of bands. It is from the dataset HDA100 [52] and is captured by the Airborne Visible InfraRed Imaging Spectrometer-Next Generation (AVIRIS-NG). The false-color image and ground truth of IM4 are shown in Fig. 5(d) and (i), respectively.

e) *IM5:* This image has a size of 64 × 64 × 276, where 276 is the number of bands. It is also from the dataset HDA100 and is captured by the AVIRIS-NG. The false-color image and ground truth of IM5 are shown in Fig. 5(e) and (j), respectively.

f) *IM6:* The image is a spectral vehicle dataset acquired by the AVIRIS sensor [53]. The spatial resolution of the image is 3.3 m. The size of the dataset is 100 × 400 × 224. IM6 targets two aircraft, whose pseudo-color images and truth maps are obtained from Fig. 6(a) and (b), respectively.

B. Experimental Settings

In the meta-training of TCFSL, samples from the type of sensor are randomly selected in the way of a two-way five-shot to form a support set and the sample number per class in the query set is 19 in each episode. For the SD, all data are used to construct a base set. For the TD, the base set consists of five target pixels selected as prior and five background pixels selected from the predetection map. Several significant parameters, namely the training episode, learning rate and spatial patch size are set as follows: episode = 10000, learning rate = 0.001, and spatial patch size = 9. λ_1 and λ_2 are set as

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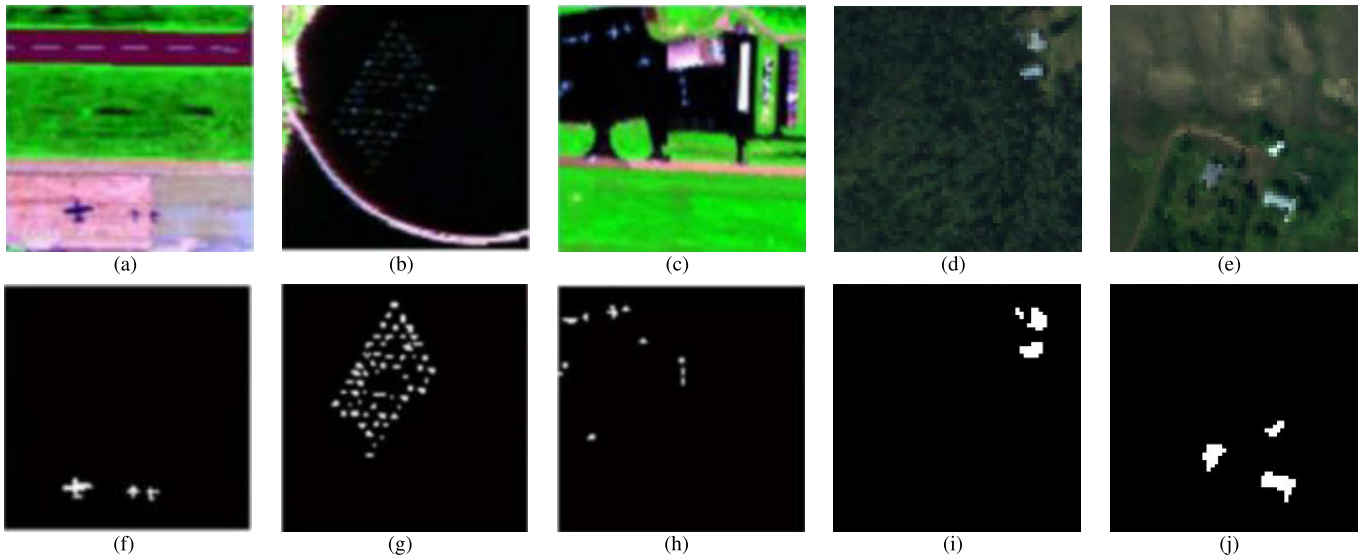


Fig. 5. False-color images and ground-truth maps. (a)–(e) False-color images of IM1–IM5. (f)–(j) Corresponding ground-truth maps.

(1e−0, 1e−1) for IM1 to IM3 and (1e−0, 1e−3) for IM4 to IM5, respectively.

For deep learning methods, random sampling is mostly done under the assumption that the target occupies a large number of pixels. The sampling will ignore the limitation that the ground truth has only a small number of labeled data samples or only subpixels.

Metrics for HTD are a 2-D receiver-operating characteristic (ROC) curve of (P_D, P_F), a 3-D ROC curve of (P_D, P_F, τ), and an area under the curve (AUC). P_D is referred to as the sensitivity or the rate of true positives and the P_F is referred to as the rate of false negatives. τ is a threshold value ranging from [0, 1].

The method proposed in this article is evaluated using 2D-ROC, 3D-ROC, and box plots. The 2D-ROC curve of (P_D, P_F) can indicate the detection effect more intuitively and the 3D-ROC curve can display the target detection ability and background suppression ability more comprehensively. In the 2D-ROC curve, the closer the ROC curve is to the upper left space of the theoretical ROC space, the better detection accuracy is obtained. In the 3D-ROC curve, the closer the ROC curve in the (P_D, P_F) profile is to the upper left corner, the better the target detection efficiency. The closer the ROC curve of the (P_F, τ) profile is to the lower left corner, the higher the background suppression ability. The closer the ROC curve of the (P_D, τ) profile is to the upper left corner, the better the object detection capability [54]. Thus, the 3D-ROC curve shows detection effectiveness more comprehensively. The AUC is the integral under the ROC curve. The closer the AUC to 1, the better the detection result indicates. With the above three two-dimensional ROC curves and their corresponding AUC, the background suppression AUC_{BS}, target detectability in the background $\text{AUC}_{\text{TD-BS}}$, and signal-to-noise ratio probability AUC_{SNPR} can be calculated. Box plots are mainly used to reflect the distribution characteristics of raw data and to compare the distribution characteristics of multiple sets of data. In hyperspectral object detection, it is a visualization tool used to evaluate the separability of targets and backgrounds. When drawing box plots, five values from a set of data are conventionally used; the extremes, the upper and lower quartiles, and the median. Box plots usually also include "whiskers," lines that extend from the ends of the box to indicate the extent of the data [55].

With regard to the hardware equipment, traditional detection methods are implemented on a CPU consisting of Intel[1] Core[2] i5-7200U CPU @ 2.50 GHz with a RAM of 8 GB, and the represent-based and deep learning-based methods are carried out on a computer consisting of an Intel Xeon[1] Sliver 4210 processor with 10M of cache and an NVIDIA GeForce GTX 2080Ti GPU.

C. Parameter Analysis and Ablation Study

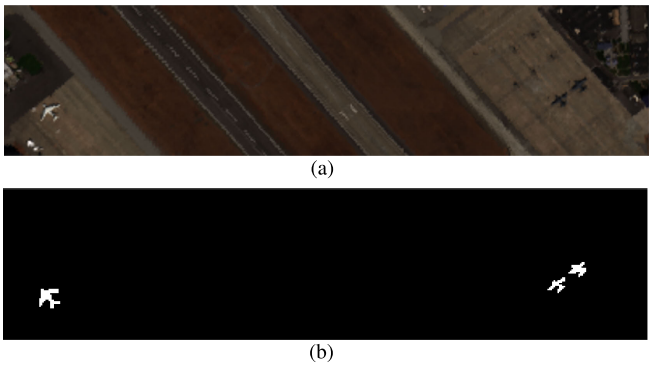
Two parameters λ_1 and λ_2 have significant impacts on the detection effect, so an analysis of them is necessary. The regularization parameters, that is, λ_1 and λ_2, regarded as adjustable hyperparameters, are selected from $\{1e-2, 1e-1, 1e-0, 1e+1\}$ and $\{1e-3, 1e-2, 1e-1\}$, respectively. The variation trend of the AUC under different combinations of λ_1 and λ_2 values on three testing HSIs are presented in Table I.

It can be noted that different datasets achieve the optimization under different combinations of λ_1 and λ_2. This is because different datasets with different coverings have different distributions, so feature-level domain alignment and distribution-level domain alignment put diverse weights.

The proposed framework mainly contains three parts: the cross-domain FSL block and feature-level and distribution-level DA blocks. To assess the contribution of key components of the TCFSL and verify the effectiveness of each block in the network, ablation analyses are performed and analyzed by removing or modifying each component from this proposed framework.

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(a) (b)

Fig. 6. False color images and ground-truth maps of IM6. (a) False-color image. (b) Ground-truth map.

There are four components in the ablation analyses: 1) T-FSL: the TD FSL module; 2) S-FSL: the SD FSL module; 3) F-DA: feature-level DA block; and 4) D-DA: distribution-level adaptation block; The variation trend of AUC under different components on three testing HSIs is presented in Table II. * in Table II denotes that the cross-domain FSL and the two DA blocks are trained phased-by-phased instead of training uniformly.

It can be seen from the table the following.

1) For all datasets, only using TD data for FSL (T-FSL) results in unsatisfied detection results due to limited training samples.
2) For all datasets, only using SD data for FSL (S-FSL) also presents discontented detection accuracy due to the great distribution discrepancy between SD and TD.
3) When cross-domain FSL is presented (T-FSL combined with S-FSL, namely cross-domain FSL), detection effects are improved on IM1, IM3, IM4, and IM6, while declined on IM2 and IM5, which demonstrated the interference of domain shift.
4) When the feature-level DA module is added (cross-domain FSL and F-DA), detection results on IM1–IM5 datasets are further prompted compared with TCFSL, which shows that the intradomain feature-level information is important.
5) When the distribution-level DA block is used, while the feature-level DA block is removed (cross-domain FSL and D-DA), the AUC values on IM1, IM2, and IM5 are improved although they decline on IM3 and IM4. It demonstrates that distribution-level DA can facilitate the detection ability to some extent, while a relatively small weight is properly compared with DA, which is why the ratio of λ_1 is greater than λ_2. The facilitation of distribution-level DA also shows the effect of the cross-ViT. It is the cross-ViT that captures distribution-level information of two data domains, then through the domain classifier obtaining the distribution-level similarities and extracting useful information, which prompts the detection effectiveness.
6) When all the proposed blocks are used but trained segmented, AUC declines significantly to a different extent on five datasets. It demonstrated that it is necessary to train the whole network unitary to realize better information interaction and parameter adjustment.
7) When all the proposed blocks are applied, the detection results on five datasets achieve the best effects, which manifest the effectiveness of each component of our proposed method.

TABLE I
AUC VALUES ON SIX DATASETS BASED ON DIFFERENT PARAMETERS

Data sets	IM1	IM2	IM3	IM4	IM5	IM6
(1e-2, 1e-3)	0.99119	0.98461	0.99572	0.99980	0.98401	0.83137
(1e-1, 1e-3)	0.99579	0.95544	0.99716	0.99980	0.98387	0.85322
(1e-0, 1e-3)	0.99057	0.98096	**0.99726**	0.99999	**0.99320**	0.85133
(1e+1, 1e-3)	**0.99694**	0.99115	0.98806	0.99994	0.95326	0.80503
(1e-2, 1e-2)	0.99129	0.98555	0.99671	0.99516	0.98932	0.81831
(1e-1, 1e-2)	0.97825	0.97660	0.99470	**1.00000**	0.98231	0.84102
(1e-0, 1e-2)	0.99326	0.95861	0.99715	0.99998	0.98838	0.84056
(1e+1, 1e-2)	0.98868	0.97814	0.99593	0.99991	0.95029	0.81857
(1e-2, 1e-1)	0.99068	0.97443	0.98599	0.99927	0.96349	0.84615
(1e-1, 1e-1)	0.98469	0.96602	0.98698	0.99913	0.99132	0.82304
(1e-0, 1e-1)	0.99555	**0.99362**	0.99684	0.99998	0.98525	**0.85423**
(1e+1, 1e-1)	0.97861	0.97776	0.99324	0.99917	0.98116	0.82745

TABLE II
AUC VALUES ON SIX DATASETS BASED ON DIFFERENT COMBINATIONS OF COMPONENTS

T-FSL	✓		✓	✓	✓	✓*	✓
S-FSL		✓	✓	✓	✓	✓*	✓
F-DA				✓		✓*	✓
D-DA					✓	✓*	✓
IM1	0.98151	0.91760	0.99426	0.99659	0.99555	0.85849	0.99555
IM2	0.97398	0.97664	0.97159	0.98674	0.97929	0.96659	0.99362
IM3	0.98263	0.94782	0.99596	0.99650	0.98801	0.94710	0.99684
IM4	0.99991	0.98096	0.99982	0.99992	0.99679	0.89217	0.99998
IM5	0.93232	0.97304	0.96769	0.98559	0.98881	0.86277	0.98525
IM6	0.82578	0.61127	0.84418	0.82042	0.86373	0.58331	0.85423

D. Comparison With State-of-the-Art Methods

In this section, the target detection performance of the proposed TCFSL is evaluated. We select twelve compared methods, including two classic algorithms CEM [15] and LSOSP [19], five improved algorithms SPSMF [21], CTTD [26], hCEM [23], E-CEM [24], and HySRTD [27], a representation-based algorithm DSC [31], and three deep learning-based algorithms HTD-IRN [39], AMSSD [38], and MLSN [46], in which the CNNTD and the MLSN are based on transfer learning. In traditional methods and two deep learning-based MLSN and HTD-IRN, only one target prior pixel is selected randomly as target information. In AMMSD and the proposed TCFSL, several target prior pixels are selected arbitrarily as prior target information. Parameters in DSC are in accordance with the original publication. In MLSN, the number of iterations is set to 60 000, and the epoch, batch size, and learning rate are set to 5, 1000, and 0.001, respectively. In AMSSD, 15% targets and 15%

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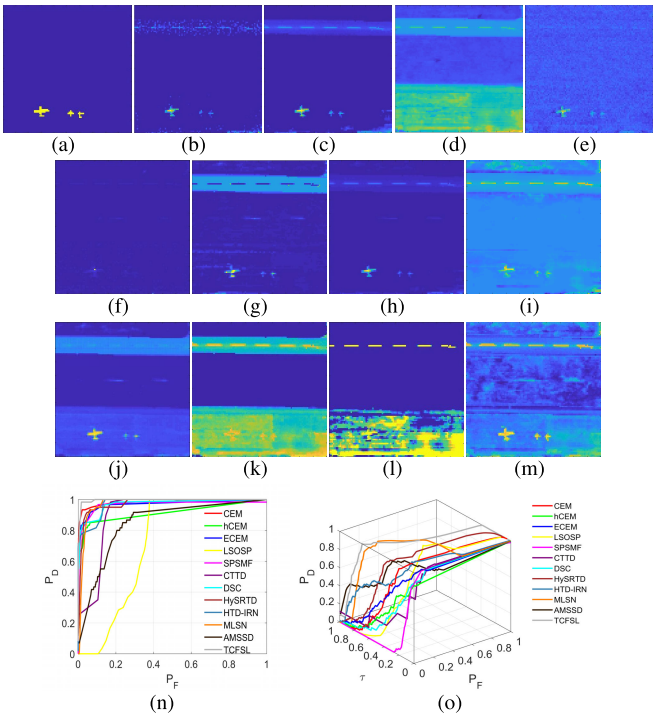


Fig. 7. Results under different methods on IM1. (a) Ground truth. (b) CEM. (c) hCEM. (d) E-CEM. (e) LSOSP. (f) SPSMF. (g) CTTD. (h) DSC. (i) HySRTD. (j) HTD-IRN. (k) MLSN. (l) AMSSD. (m) TCFSL. (n) 2D-ROC curve. (o) 3D-ROC curve.

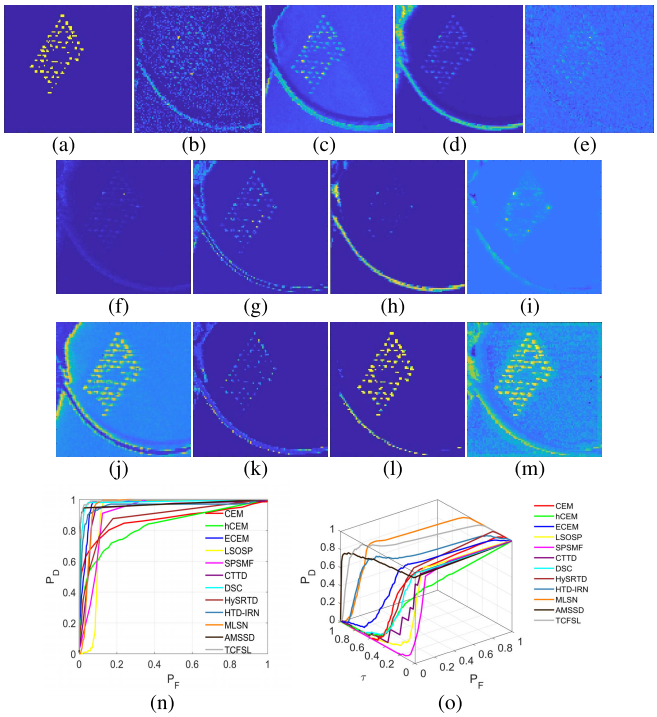


Fig. 8. Results under different methods on IM2. (a) Ground truth. (b) CEM. (c) hCEM. (d) E-CEM. (e) LSOSP. (f) SPSMF. (g) CTTD. (h) DSC. (i) HySRTD. (j) HTD-IRN. (k) MLSN. (l) AMSSD. (m) TCFSL. (n) 2D-ROC curve. (o) 3D-ROC curve.

backgrounds are selected as priori for training, and extra data augmentation is added for IM4 and IM5 because they have small scales. Hyperparameters learning rate, batch size, training epoch, and patch size are set to 0.001, 256, 20, and 3, respectively.

1) *Qualitative Results:* Figs. 7(a)–(m)–11(a)–(m) present the detection map of the 11 compared methods and the proposed TCFSL method on the three testing HSIs. For IM1, E-CEM detects very few targets and has a high false alarm rate. CEM, hCEM, SPSMF, and CTTD can detect parts of targets. LSOSP, HySRTD, HTD-Net, MLSN, and AMSSD can also detect some targets, but they have a large false alarm rate. Comparatively, DSC, HTD-IRN, and the TCFSL detect more targets and the proposed TCFLS gives a more intact detection map, and the DSC and HTD-IRN show relatively better background suppression competence.

For IM2, CEM and LSOSP can detect some target pixels with a heavy interface of noise. The E-CEM, SPSMF, CTTD, DSC, HySRTD, and MLSN can detect part targets and show great competence in background suppression. The hCEM and AMSSD also suppress background well and detect most targets. The HTD-IRN and the TCFSL detect almost all targets and show a declined ability to suppress background compared with the AMSSD.

For IM3, CEM and LSOSP can detect parts of target pixels, while noise abounds in their detection maps. hCEM, SPSMF, CTTD, DSC, and MLSN suppress background greatly, while the targets in the result of the SPSMF are not highlighted, and the MLSN has a high missed detection rate and false alarm rate. The E-CEM, the HySRTD, the AMSSD, the HTD-IRN, and the TCFSL highlight the target best, and the E-CEM, the HySRTD, the AMSSD, and the HTD-IRN have a higher false alarm rate comparatively.

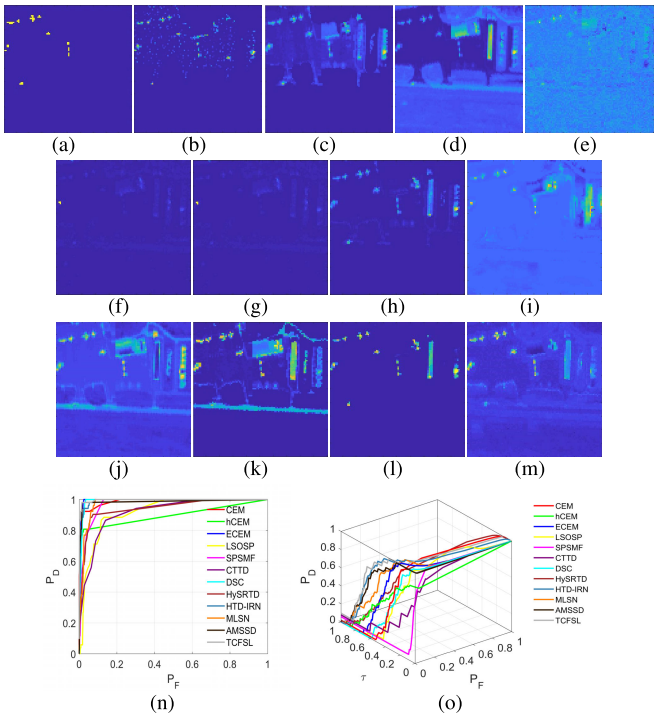


Fig. 9. Results under different methods on IM3. (a) Ground truth. (b) CEM. (c) hCEM. (d) E-CEM. (e) LSOSP. (f) SPSMF. (g) CTTD. (h) DSC. (i) HySRTD. (j) HTD-IRN. (k) MLSN. (l) AMSSD. (m) TCFSL. (n) 2D-ROC curve. (o) 3D-ROC curve.

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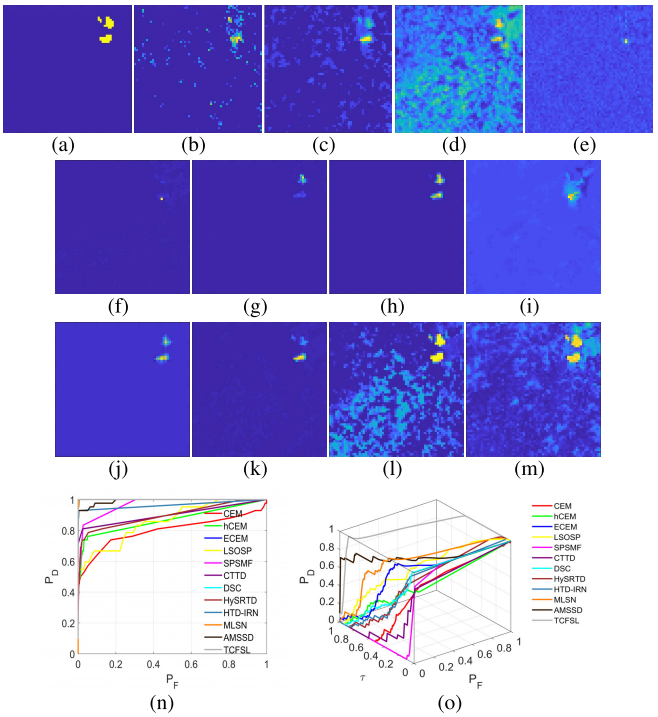


Fig. 10. Results under different methods on IM4. (a) Ground truth. (b) CEM. (c) hCEM. (d) E-CEM. (e) LSOSP. (f) SPSMF. (g) CTTD. (h) DSC. (i) HySRTD. (j) HTD-IRN. (k) MLSN. (l) AMSSD. (m) TCFSL. (n) 2D-ROC curve. (o) 3D-ROC curve.

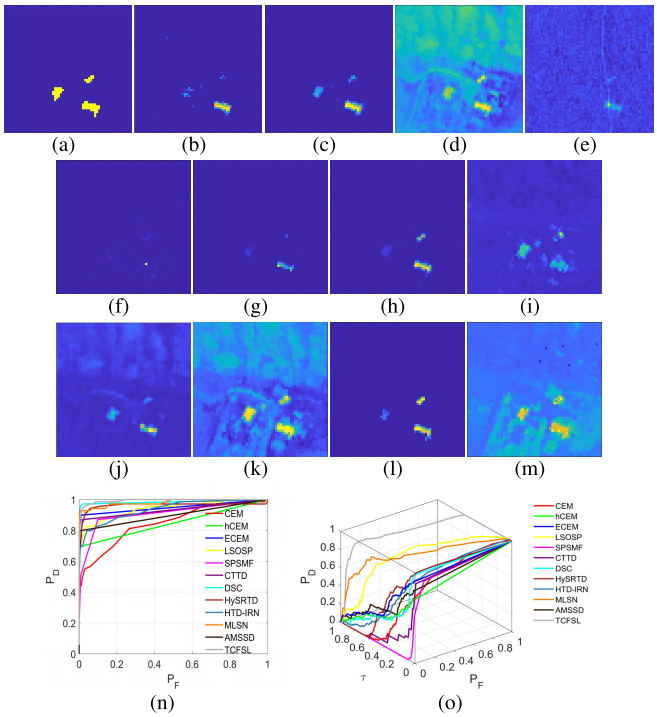


Fig. 11. Results under different methods on IM5. (a) Ground truth. (b) CEM. (c) hCEM. (d) E-CEM. (e) LSOSP. (f) SPSMF. (g) CTTD. (h) DSC. (i) HySRTD. (j) HTD-IRN. (k) MLSN. (l) AMSSD. (m) TCFSL. (n) 2D-ROC curve. (o) 3D-ROC curve.

For IM4, CEM, hCEM, E-CEM, and HySRTD can detect parts of the target with a lot of noise. The LSOSP, SPSMF, CTTD, DSC, HTD-IRN, and MLSN have great background suppression competence, while they detect incomplete maps. The AMSSD and TCFSL can extract the target and restrain background simultaneously and the TCFSL detects more target pixels.

For IM5, the E-CEM and LSOSP are capable of detecting some targets but their detection maps are contaminated with a significant amount of noise. The CEM, hCEM, SPSMF, CTTD, DSC, HySRTD, HTD-IRN, and AMSSD show great ability in background suppression while targets are neglected, especially the SPSMF. The MLSN and the TCFSL detect most targets relative to other methods and also highlight the targets more prominently, thus achieving good separation between the targets and the backgrounds.

For IM6, the detection maps of HTD-IRN and AMSSD have a large number of yellow false alarms, which can be seen as a poor detection capability. The LSOSP and CTTD have poor background suppression from the LSOSP and CTTD, and obvious background ripples can be seen on their detection maps. The hCEM, E-CEM, SPSMF, HySRTD, and MLSN have better background suppression, but they are poor at detecting targets, for example, SPSMF and HySRTD did not detect the two aircraft in the lower right corner. It can be seen that CEM, DSC, and TCFSL detect more complete targets. Among them, CEM and TCFSL have poor noise suppression, DSC has good noise suppression but poor background suppression, and TCFSL has more complete detection of the three targets.

2) *Quantitative Evaluation and Analysis:* Intuitionistic evaluation of algorithm performances just by detection map makes it difficult to decide the detection effects. To evaluate the detection performance of each algorithm more accurately and comprehensively, the 2D-ROC curve, 3D-ROC curve, and AUC were used for quantitative evaluation. The box-plot figure is also used to further estimate the ability of different methods on target–background separability. The 2D-ROC curves and box-plot figures of all methods on different datasets are displayed in Figs. 7(m) and (n)–12(m) and (n) and Fig. 13, respectively, and the AUC values are listed in Table III. For the 2D-ROC evaluator, the 2D-ROC curves of the TCFSL get close to the top left corner. For the 3D-ROC curve, it can be seen that the TCFSL reaches better target detection ability and has relatively high background suppression ability compared with other methods. For the AUC, the AUC values of the TCFSL reach the best values on six testing HSIs. For the target detection efficiency $\mathrm{AUC}_{(D,F)}$, the AUC value of TCFSL reaches its optimal value on six test HSIs. At the same time, the $\mathrm{AUC}_{(D,\tau)}$ of TCFSL is also the maximum value on five datasets. Combined with the above result images, it can be seen that TCFSL has excellent target detection ability and can always detect target pixels completely and comprehensively. From $\mathrm{AUC}_{(F,\tau)}$ and $\mathrm{AUC}_{\mathrm{BS}}$, it can be seen that TCFSL has weak background suppression ability, which is also reflected in $\mathrm{AUC}_{\mathrm{TD\text{-}BS}}$. TCFSL has a strong ability in object detection, but due to the lack of particularly excellent background suppression, the evaluation of object detection ability in the background is limited. Finally, the $\mathrm{AUC}_{\mathrm{SNPR}}$ of TCFSL is at a moderate level among various comparison algorithms.

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TABLE III
AUC VALUES OF ALL METHODS ON FIVE DATASETS

Datasets	AUCs	Detector											
		CEM	hCEM	E-CEM	LSOSP	SPSMF	CTTD	DSC	HySRTD	HTD-IRN	MLSN	AMSSD	TCFSL
IM1	$AUC_{(D,F)}$	0.98429	0.92070	0.98335	0.71506	0.98702	0.91027	0.97556	0.97364	0.97510	0.97113	0.87505	**0.99555**
	$AUC_{(D,\tau)}$	0.42983	0.31950	0.43400	0.49033	0.10675	0.32933	0.28550	0.58250	0.62900	0.78350	0.69283	**0.84367**
	$AUC_{(F,\tau)}$	0.16709	**0.00631**	0.01995	0.31976	0.02627	0.05745	0.01056	0.31361	0.09754	0.29059	0.18910	0.48592
	AUC_{BS}	0.81720	0.91439	0.96340	0.39530	0.96075	0.85282	**0.96500**	0.66003	0.87756	0.68054	0.68595	0.50963
	AUC_{TD-BS}	0.26274	0.31319	0.41405	0.17057	0.08048	0.27188	0.27494	0.26889	**0.53146**	0.49291	0.50373	0.35775
	AUC_{SNPR}	2.57245	**50.63391**	21.75439	1.53343	4.06357	5.73246	27.03598	1.85740	6.44864	2.69624	3.66383	1.73623
IM2	$AUC_{(D,F)}$	0.86709	0.82860	0.96911	0.90273	0.92267	0.92951	0.97587	0.88398	0.95546	0.95197	0.99226	**0.99362**
	$AUC_{(D,\tau)}$	0.31728	0.23755	0.45619	0.13941	0.04376	0.19119	0.25111	0.32792	0.74426	0.80886	0.86584	**0.89980**
	$AUC_{(F,\tau)}$	0.21714	0.04726	0.14534	0.06227	0.01209	0.03415	**0.00985**	0.25078	0.30734	0.64542	0.01479	0.50102
	AUC_{BS}	0.64995	0.78134	0.82377	0.84046	0.91058	0.89536	0.96602	0.63320	0.64812	0.30655	**0.97747**	0.49260
	AUC_{TD-BS}	0.10014	0.19029	0.31085	0.07714	0.03167	0.15704	0.24126	0.07714	0.43692	0.16344	**0.85105**	0.39878
	AUC_{SNPR}	1.46118	5.02645	3.13878	2.23880	3.61952	5.59854	25.49340	1.30760	2.42162	1.25323	**58.54226**	1.79594
IM3	$AUC_{(D,F)}$	0.98809	0.89573	0.99414	0.91933	0.97132	0.91783	0.99046	0.94350	0.99077	0.97607	0.99480	**0.99684**
	$AUC_{(D,\tau)}$	0.34181	0.40846	0.47269	0.32558	0.09789	0.25058	0.33115	0.49058	0.67404	0.60481	0.64327	**0.69942**
	$AUC_{(F,\tau)}$	0.17382	**0.00703**	0.03388	0.13091	0.01367	0.02861	0.01383	0.21906	0.12938	0.05762	0.01416	0.15324
	AUC_{BS}	0.81427	0.88870	0.96026	0.78842	0.95765	0.88922	0.97663	0.72444	0.86139	0.91845	**0.98064**	0.84360
	AUC_{TD-BS}	0.16799	0.40143	0.43881	0.19467	0.08422	0.22197	0.31732	0.27152	0.54466	0.54719	**0.62911**	0.54618
	AUC_{SNPR}	1.96646	**58.10242**	13.95189	2.48705	7.16094	8.75848	23.94432	2.23948	5.20977	10.49653	45.42867	4.56421
IM4	$AUC_{(D,F)}$	0.80068	0.87070	0.99990	0.86138	0.97965	0.97017	0.99997	0.89990	0.99935	0.99970	0.99235	**0.99999**
	$AUC_{(D,\tau)}$	0.24209	0.41430	0.53965	0.60174	0.05779	0.15430	0.40663	0.37221	0.36174	0.68477	0.89337	**0.93942**
	$AUC_{(F,\tau)}$	0.11457	0.01132	0.02461	0.28595	0.00835	0.00477	**0.00435**	0.13678	0.05509	0.18296	0.09757	0.71296
	AUC_{BS}	0.68611	0.85938	0.97529	0.57543	0.97130	0.9654	**0.99562**	0.76312	0.94426	0.81674	0.89478	0.28703
	AUC_{TD-BS}	0.12752	0.40298	0.51504	0.31579	0.04944	0.14953	0.40228	0.23543	0.30665	0.50181	**0.79580**	0.22646
	AUC_{SNPR}	2.11303	36.59894	21.92808	2.10435	6.92096	32.34801	**93.47816**	2.72123	6.56635	3.74273	9.15620	1.31763
IM5	$AUC_{(D,F)}$	0.84630	0.84837	0.97708	0.94303	0.96475	0.94360	0.98331	0.96283	0.93337	0.98132	0.97180	**0.99320**
	$AUC_{(D,\tau)}$	0.23300	0.31800	0.41514	0.70057	0.03729	0.12300	0.31164	0.36557	0.35643	0.80843	0.38629	**0.91386**
	$AUC_{(F,\tau)}$	0.10267	0.00572	0.00625	0.37592	0.00607	0.00517	0.00453	0.05111	0.07101	0.24633	**0.00121**	0.69374
	AUC_{BS}	0.74363	0.84265	0.97083	0.56711	0.95868	0.93843	**0.97878**	0.91172	0.86236	0.73499	0.97059	0.29946
	AUC_{TD-BS}	0.13033	0.31228	0.40889	0.32465	0.03122	0.11783	0.30711	0.31446	0.28542	**0.56210**	0.38508	0.22012
	AUC_{SNPR}	2.26941	55.59441	66.42240	1.86361	6.14333	23.79110	68.79470	7.15261	5.01943	3.28190	**319.24793**	1.31729
IM6	$AUC_{(D,F)}$	0.76043	0.67813	0.68353	0.50200	0.69327	0.55917	0.78333	0.77595	0.62821	0.55932	0.60463	**0.85423**
	$AUC_{(D,\tau)}$	0.51345	0.16810	0.20256	0.26414	0.01345	0.27142	0.24074	0.22604	**0.75612**	0.07911	0.28330	0.27497
	$AUC_{(F,\tau)}$	0.42262	0.06033	0.06393	0.21747	**0.00544**	0.13274	0.02396	0.08173	0.61585	0.05938	0.11170	0.12204
	AUC_{BS}	0.33781	0.61780	0.61960	0.28453	0.68783	0.42643	**0.75937**	0.69422	0.01236	0.49994	0.49293	0.73219
	AUC_{TD-BS}	0.09083	0.10777	0.13863	0.04667	0.00801	0.13868	**0.21678**	0.14431	0.14027	0.01973	0.17160	0.15293
	AUC_{SNPR}	1.21492	2.78634	3.16827	1.21460	2.47107	2.04475	**10.04842**	2.76562	1.22777	1.33226	2.53626	2.25311

As for the box-plot figure, it can be seen from Fig. 13 that the TCFSL achieves a relatively better separability between the background and the target compared with other methods. In conclusion, the TCFSL presents a better detection competence than the eleven compared classic and start-of-the-art methods on six real testing HSIs. The box length of the box plot we use in Fig. 13 is [25%, 75%] because this range is widely used in the field of statistics. In particular, in the field of hyperspectral detection, it has also been suggested that a range of [10%, 90%] is consistent with empirical choices when

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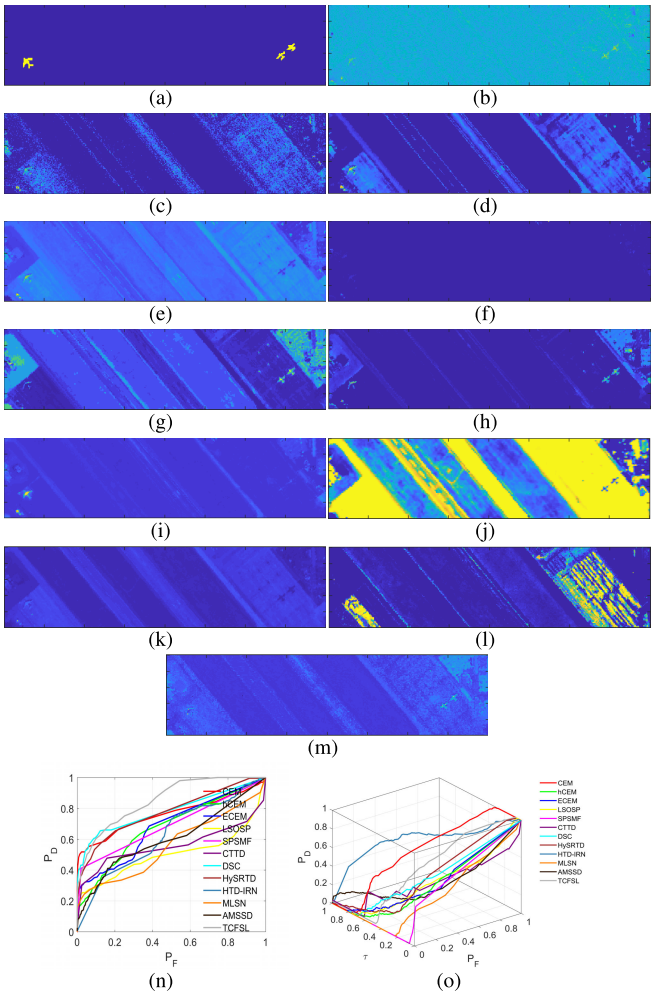


Fig. 12. Results under different methods on IM6. (a) Ground truth. (b) CEM. (c) hCEM. (d) E-CEM. (e) LSOSP. (f) SPSMF. (g) CTTD. (h) DSC. (i) HySRTD. (j) HTD-IRN. (k) MLSN. (l) AMSSD. (m) TCFSL. (n) 2D-ROC curve. (o) 3D-ROC curve.

using box plots for data analysis [56], [57]. Therefore, a box plot with a range of [10%, 90%] was used to analyze the data for IM1 as shown in Fig. 14. It can be seen that on IM1, TCFSL can separate target and background better than other algorithms.

In summary, TCFSL shows better results in both quantitative and qualitative analyses, which is a result of the effectiveness of the mechanism proposed in this article. First, the cross-domain few-shot used by TCFSL effectively solves the problem of limited a priori information in HTD. Obtaining more comprehensive information allows the model to fully extract the features of the target during training and detect a more complete target during testing. From the detection maps of IM1–IM6, it can be seen that the TCFSL proposed in this article always has a better target highlighting effect, which can obtain a more complete target detection image and achieve better target and background separation. In addition, as can be seen from Table III, the $\mathrm{AUC}_{(D,\tau)}$ value of TCFSL is the largest on all five datasets, which is a reflection of the target detection ability. Second, TCFSL's overall optimization of FSL and DA, together with the use of DA on both scales to solve the domain bias problem, may also make the detection results more accurate. It can be seen in the detection plots of IM1–IM6 that TCFSL hardly misses correct detection of targets, and it has the highest $\mathrm{AUC}_{(D,F)}$ on every dataset in Table III.

TABLE IV
COMPUTATIONAL TIME (IN SECONDS) OF ALL METHODS ON FIVE DATASETS

Algorithm	Time	IM1	IM2	IM3	IM4	IM5	IM6
CEM	Total	0.06	0.06	0.04	0.07	0.05	0.29
hCEM	Total	2.93	5.86	2.63	1.97	4.44	5.91
E-CEM	Total	4.69	3.94	3.82	3.42	3.33	22.10
LSOSP	Total	25.23	30.86	26.72	1.11	1.10	695.10
SPSMF	Total	0.45	0.42	0.34	0.281	0.28	1.91
CTTD	Total	0.02	0.03	0.02	0.03	0.03	0.34
DSC	Total	7.34	12.21	15.94	9.70	62.17	116.41
HySRTD	Total	78.20	99.06	80.56	29.15	29.15	359.30
HTD-IRN	Train	221.26	226.61	222.57	211.14	210.28	770.48
	Test	0.83	0.84	0.81	0.78	0.77	5.95
MLSN	Train	1364.01	1364.01	1364.01	1115.41	1115.44	12808.92
	Test	18.24	18.50	131.54	7.58	7.470	90.81
AMSSD	Train	265.61	135.45	248.62	196.50	232.70	183.18
	Test	215.65	91.19	222.16	38.44	38.71	619.11
TCFSL	Train	2817.75	2717.16	3328.44	2766.68	2742.67	4055.17
	Test	2.17	2.32	2.57	0.96	1.03	14.61

E. Analysis of Computational Time

In this section, the time cost of the proposed method is evaluated. To compare the computational time of different methods, all the experiments are implemented on the same computer. The execution time of compared classic and several state-of-art methods are presented in Table IV.

It can be seen from Table IV that traditional methods are less time-consuming due to their simple algorithmic structure. On the contrary, the deep-learning-based methods cost the most time because of their huge parameters. Compared with the CNNTD, the AMSSD costs most of the time because it processes patches instead of pixels in CNNTD and it uses a more complicated network. Compared with methods without cross-domain, the MLSN and the proposed TCFSL consume the most time on training, which indicates that the computational cost of transfer learning is more expensive. The MLSN costs less training, this is because the TCFSL processes spectral and spatial information simultaneously, while the MLSN just extracts spectral features. Compared with other deep-learning methods, the TCFSL has less testing time cost, though it has a relatively longer training time. Thus, the computing time for TCFSL is overall acceptable.

V. CONCLUSION

In this article, a TCFSL method was proposed oriented toward the issues of the limited target priori and class imbal-

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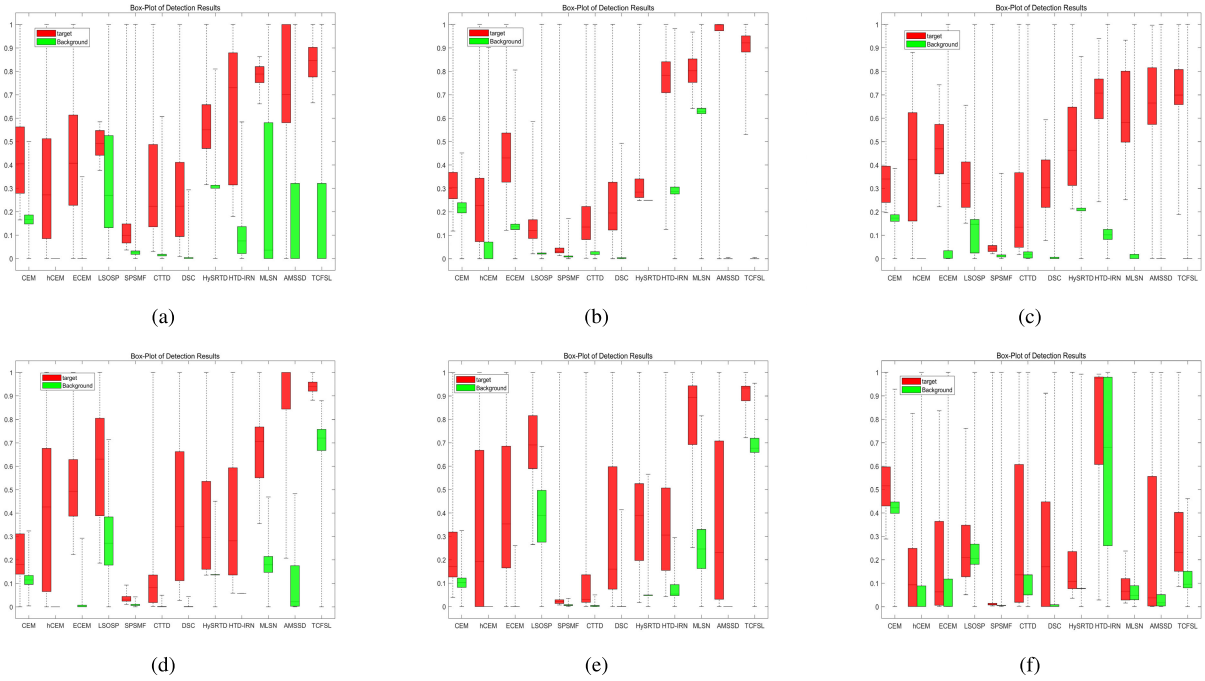


Fig. 13. Box-plot figures of IM1–IM6 datasets.

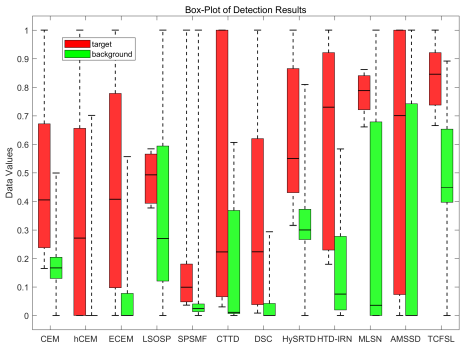


Fig. 14. Box plot of IM1 in the [10%, 90%] range.

ance in HTD. The proposed method devised a cross-domain FSL framework, which can learn transferable knowledge in SD, and constructed a discriminative discriminator for TD to achieve target detection in TD with few labeled samples. To alleviate the interference of domain shift in the cross-domain FSL, feature-level and domain-level DA are both applied to excavate domain-invariant features and interdomain distribution-level information, to facilitate FSL tasks in each data domain and improve target detection accuracy. To prompt information interaction between FSL and DA, the DA block is unified with the cross-domain FSL block to present feature interaction and parameter adjustment more effectively, to obtain the optimal model. Experimental results indicate that the proposed method outperforms the compared classic and deep learning-based methods on five datasets.

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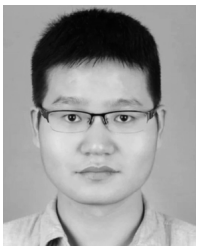
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